Change Detection Over a Decade in Southern Ontario Using Remotely Sensed Landsat Data (2014–2024)

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7th April 2026

Introduction

Metropolitan cities and their suburbs are always expanding, making planning essential. This report presents the results of changes that have occurred in the Southern part of Ontario, mainly Toronto and its surrounding cities, such as Milton, Mississauga, Brampton, Bolton, Vaughan and stretching all the way to Ajax in the east.

The analysis uses two atmospherically corrected Landsat 8 scenes collected on 19 August 2014 and 14 August 2024. Clouds and shadows have been masked accordingly to avoid misinterpretation of data. Each Landsat scene covers path 18 / row 30 and contains eight multispectral bands at 30 m spatial resolution, although we have only used bands 2 to 7 for this project. The Pixel Quality Assessment band provides information on cloud cover and shadows.

The report will examine the change-detection techniques used, their results and limitations in urban planning.

Methodology

The entire pipeline was created and processed using Whitebox Workflows within Python notebooks. The created images were further analyzed on QGIS. First, an RGB colour-composite image was created to visually detect differences between the two data sets. A binary cloud mask was created by using the Pixel Quality Assessment band to identify clear land. This mask was applied to all change detection techniques.

Three change detection techniques were applied, namely NDVI, Image Regression, and Random Forest Classification (post-classification comparison).

1) Technique 1 – Normalized Difference Vegetation Index (NDVI)

NDVI compares near-infrared and red reflectance to characterize vegetation greenness and is calculated as *NDVI = (NIR – RED)/(NIR + RED)*
(NASA Earth Observatory, n.d). Bands 4 (NIR) and 5 (Red) are used for computation for each date. The 2014 image is then subtracted from the 2024 image; negative differences indicate vegetation loss, most likely caused by urban expansion. Positive differences, on the other hand, indicate vegetation growth.

```python
from whitebox_workflows import WbEnvironment

try:

    qa_2014 = wbe.read_raster(p2014 + "LC08_L2SP_018030_20140819_20200911_02_T1_QA_PIXEL_clipped.TIF")
    qa_2024 = wbe.read_raster(p2024 + "LC08_L2SP_018030_20240814_20240822_02_T1_QA_PIXEL_clipped.TIF")

    mask_2014 = qa_2014 == 21824
    mask_2024 = qa_2024 == 21824

    mask = mask_2014 * mask_2024

    wbe.write_raster(mask, path + "mask.tif", compress=True)

except Exception as e:
    print(f"An error occurred: {e}")
```

```python
# NDVI
ndvi_2014 = wbe.normalized_difference_index(b5_2014, b4_2014)
ndvi_2024 = wbe.normalized_difference_index(b5_2024, b4_2024)

# Masks
mask = mask_2014 * mask_2024

# Apply mask
ndvi_2014 = ndvi_2014 * mask_2014
ndvi_2024 = ndvi_2024 * mask_2024

# NDVI change
ndvi_change_2 = (ndvi_2024 - ndvi_2014) * mask

# Save
wbe.write_raster(ndvi_change_2, "ndvi_change_2.tif")
```

Figure 1. Cloud masking using QA_PIXEL values (21824) and computation of NDVI for 2014 and 2024, followed by NDVI differencing to generate a vegetation change raster.

2) Technique 2 – Image Regression

A linear regression was fitted between the 2014 (independent) and 2024 (dependent) NDVI values to model their relationship (Jensen, 2016). The regression predicts 2024 values using 2014 data, and the residuals (observed NDVI – predicted NDVI) show deviations from this trend. Residuals exceeding 2.5 standard deviations were identified as significant changes, then smoothed using a 3 × 3 median filter. The residuals were further converted to vector polygons, and only the ones with an area of more than 20,000 were retained, since the smaller polygons were pretty insignificant and merely noise. Only negative residuals (loss of greenness) are interpreted as urban expansion.

```python
# write ndvi rastersfor visual inspection
wbe.write_raster(ndvi_2014, "ndvi_2014.tif")
wbe.write_raster(ndvi_2024, "ndvi_2024.tif")

#perform image regression
regression_residuals = wbe.image_regression(
    independent_variable = ndvi_2014,
    dependent_variable = ndvi_2024,
    output_html_file = "regression_results.html",
    standardize_residuals = False,
    output_scattergram = False,
    num_samples= 10000)

regression_residuals = (abs(regression_residuals) > 0.0758*2.5) * regression_residuals
wbe.write_raster(regression_residuals, "regression_residuals.tif")
```

```python
# Apply median filter to regression residuals

median_size = 3  #3x3 filter

regression_residuals_filtered = wbe.median_filter(
    raster=regression_residuals,
    filter_size_x=median_size,
    filter_size_y=median_size
)

# Save output
wbe.write_raster(
    regression_residuals_filtered,
    path + "regression_residuals_filtered.tif",
    compress=True
)
```

```python
# Urban expansion from filtered residuals
urban_expansion = regression_residuals_filtered < 0.0

# Convert to polygons
urban_expansion_polys = wbe.raster_to_vector_polygons(urban_expansion)

# Save
wbe.write_vector(urban_expansion_polys, "urban_expansion.shp")
```

Figure 2. NDVI-based image regression, applying a residual threshold (2.5× standard deviation), smoothing results using a 3×3 median filter, and extracting urban expansion areas as vector polygons.

3) Technique 3 – Random Forest Classification

Post-classification comparison tends to be more accurate than simple image differencing (Mas, 1999). A supervised random-forest classifier was trained on 2014 data using six classes (forest, industrial/commercial, crops, residential, soil and water) and bands 2 to 7. The training data vector polygons were created on QGIS and saved as a shapefile. Each class had approximately 10 data samples each. The 100-tree model was then trained and applied to both dates. A 5x5 majority filter is then applied to reduce noise and smooth the image. A cross-tabulation matrix and a change map were then created to detect expansion in the various classes.

```python
try:
    wbe.max_procs = -1

    # Read the 7 bands of data into a variable named 'band_data'
    print('Reading band data...')
    band_data_2014 = wbe.read_rasters(
     p2014 + "LC08_L2SP_018030_20140819_20200911_02_T1_SR_B2_clipped.TIF",
     p2014 + "LC08_L2SP_018030_20140819_20200911_02_T1_SR_B3_clipped.TIF",
     p2014 + "LC08_L2SP_018030_20140819_20200911_02_T1_SR_B4_clipped.TIF",
     p2014 + "LC08_L2SP_018030_20140819_20200911_02_T1_SR_B5_clipped.TIF",
     p2014 + "LC08_L2SP_018030_20140819_20200911_02_T1_SR_B6_clipped.TIF",
     p2014 + "LC08_L2SP_018030_20140819_20200911_02_T1_SR_B7_clipped.TIF"
    )

    # Read in the training data
    print('Reading training data...')
    training_data = wbe.read_vector(p_training + 'training_2014.shp')

    # Train the model
    model_2014 = wbe.random_forest_classification_fit(
        band_data_2014,
        training_data,
        class_field_name = 'Class',
        split_criterion = "Gini",
        n_trees = 100,
        min_samples_leaf = 1,
        min_samples_split = 2,
        test_proportion = 0.2
    )

    # Use the model to predict
    rf_class_image_2014 = wbe.random_forest_classification_predict(band_data_2014, model_2014)

    # Save the output raster to disc
    wbe.write_raster(rf_class_image_2014, path + 'rf_classification_2014.tif', compress=True)

    print("Operation complete!") # Provide some sort of indication that the job is done.

except Exception as e:
    print(f"Exception: {e}")

finally:
    print(wbe.check_in_license(license_id)) # Check your license back in.
```

Figure 3. Implementation of Random Forest classification, including loading multispectral bands, importing training data, fitting the model using the Gini criterion, and generating a classified land-cover map for 2014.

```python
license_id = 'geog3420-w26'
wbe = whitebox_workflows.WbEnvironment(license_id)

try:

    # Read raster into memory (IMPORTANT FIX)
    print('Reading RF classification raster...')
    rf_2014 = wbe.read_raster(path + 'rf_classification_2014.tif')
    rf_2024 = wbe.read_raster(path + 'rf_classification_2024.tif')

    # 5x5 filter
    print('Applying 5x5 majority filter...')
    mf_2014 = wbe.majority_filter(
        rf_2014,
        filter_size_x=5,
        filter_size_y=5
    )
    wbe.write_raster(mf_2014, path + 'rf_majority_5x5_2014.tif', compress=True)

    # 5x5 filter
    print('Applying 5x5 majority filter...')
    mf_2024 = wbe.majority_filter(
        rf_2024,
        filter_size_x=5,
        filter_size_y=5
    )
    wbe.write_raster(mf_2024, path + 'rf_majority_5x5_2024.tif', compress=True)

    print("All done!")

except Exception as e:
    print(f"Error: {e}")

finally:
    wbe.check_in_license(license_id)
```

```
Reading RF classification raster...
Applying 5x5 majority filter...
Applying 5x5 majority filter...
All done!
```

Figure 4. Applying a 5×5 majority filter to smooth Random Forest classification outputs for 2014 and 2024, reducing noise and improving spatial coherence of land-cover maps.

```python
#Read in 5x5 Majority Filtered Random Forest Classified Images
rf_2014 = wbe.read_raster(path + "rf_majority_5x5_2014.tif")
rf_2024 = wbe.read_raster(path + "rf_majority_5x5_2024.tif")

wbe.cross_tabulation(
    raster1=rf_2014,
    raster2=rf_2024,
    output_html_file=path + "change_matrix.html"
)
```

```python
rf_2014 = wbe.read_raster(path + "rf_majority_5x5_2014.tif")
rf_2024 = wbe.read_raster(path + "rf_majority_5x5_2024.tif")

change_map = rf_2014 * 10 + rf_2024

wbe.write_raster(change_map, path + "rf_change_map.tif", compress=True)
```

Figure 5. Cross-tabulation between 2014 and 2024 Random Forest classified maps and generating a land-cover change map highlighting class transitions.

Results

1) NDVI

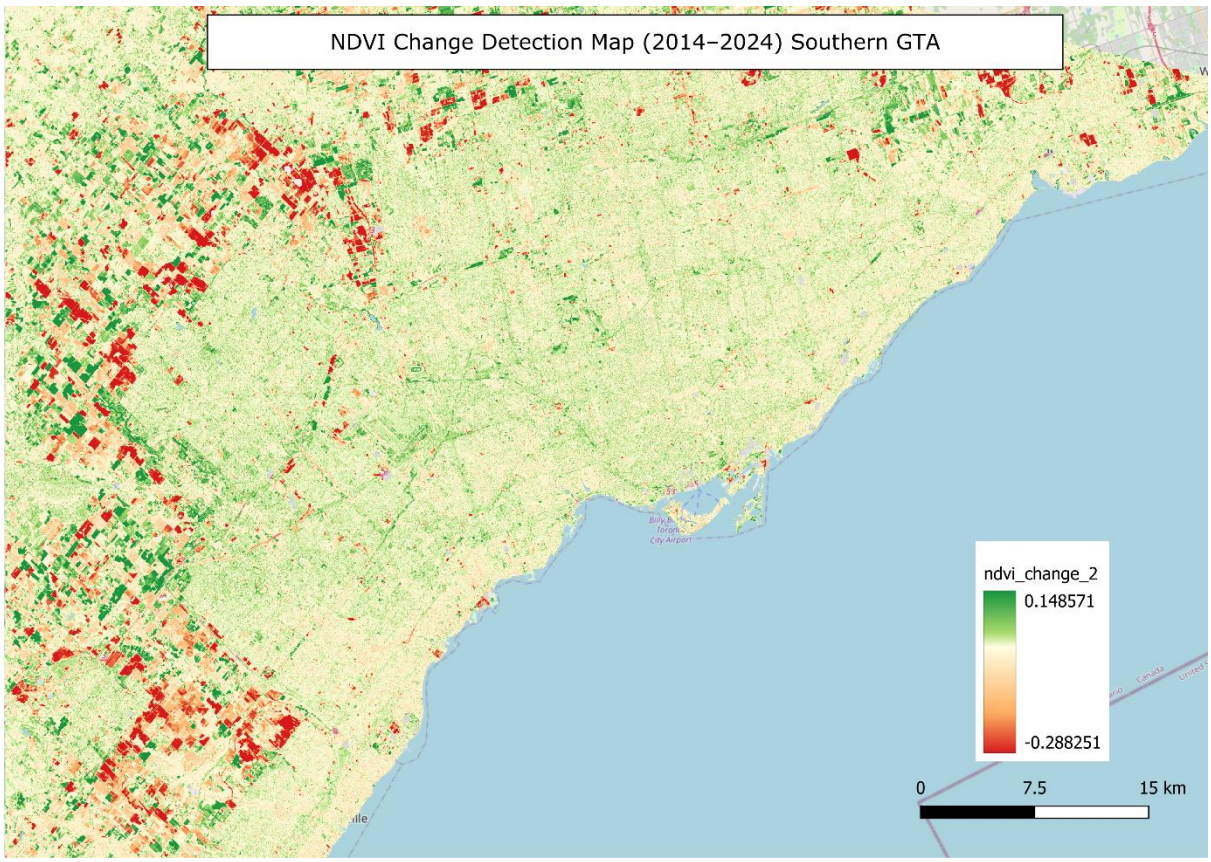

Figure 6. NDVI difference map showing changes in vegetation between 2014 and 2024. Red areas indicate vegetation loss associated with urban and industrial expansion, while green areas represent vegetation gain or crop growth. Light or neutral tones indicate little to no change. Significant expansion is visible around Milton, Brampton, and Oakville.

From the NDVI change map, the dark red patches indicate a loss of vegetation and the expansion of industrial and residential areas. The light red patches are mostly due to changes in vegetation cover or the crop being grown, and do not signify loss. The yellow or white patches indicate that there has been no change. The light green patches indicate vegetation growth, mostly on bare farmland.

Brampton and Coleraine have experienced significant urban expansion, but Oakville and Milton have changed the most over the decade. Pickering and Markham are the other two cities affected. The city of Toronto has remained the same, as there is not much room for expansion in the busy city.

2) Image Regression and Residuals

Image Regression

Input Image 1 (independent variable, X): .tif

Input Image 2 (dependent variable, Y): .tif

Model Summary

R	R Square	Std. Error of the Estimate
0.7981	0.6370	0.0758

Analysis of Variance (ANOVA)

Source	SS	df	MS	F	p
Regression	35002.0910	1.0000	35002.0910	6096008.0074	0.0000
Residual	19943.7128	3473422.0000	0.0057		
Total	54945.8038				

Coefficients

Variable	B	Std. Error	t	Sig.
Constant	0.0663	0.0001	1133.9386	0.0000
Slope	0.7338	0.0001	5592.9753	0.0000

Regression equation: .tif = 0.733763647424881 × .tif + 0.0663130279752197

Table 1. Image regression results comparing NDVI values between 2014 and 2024, used to model expected vegetation relationships and identify significant deviations through residual analysis.

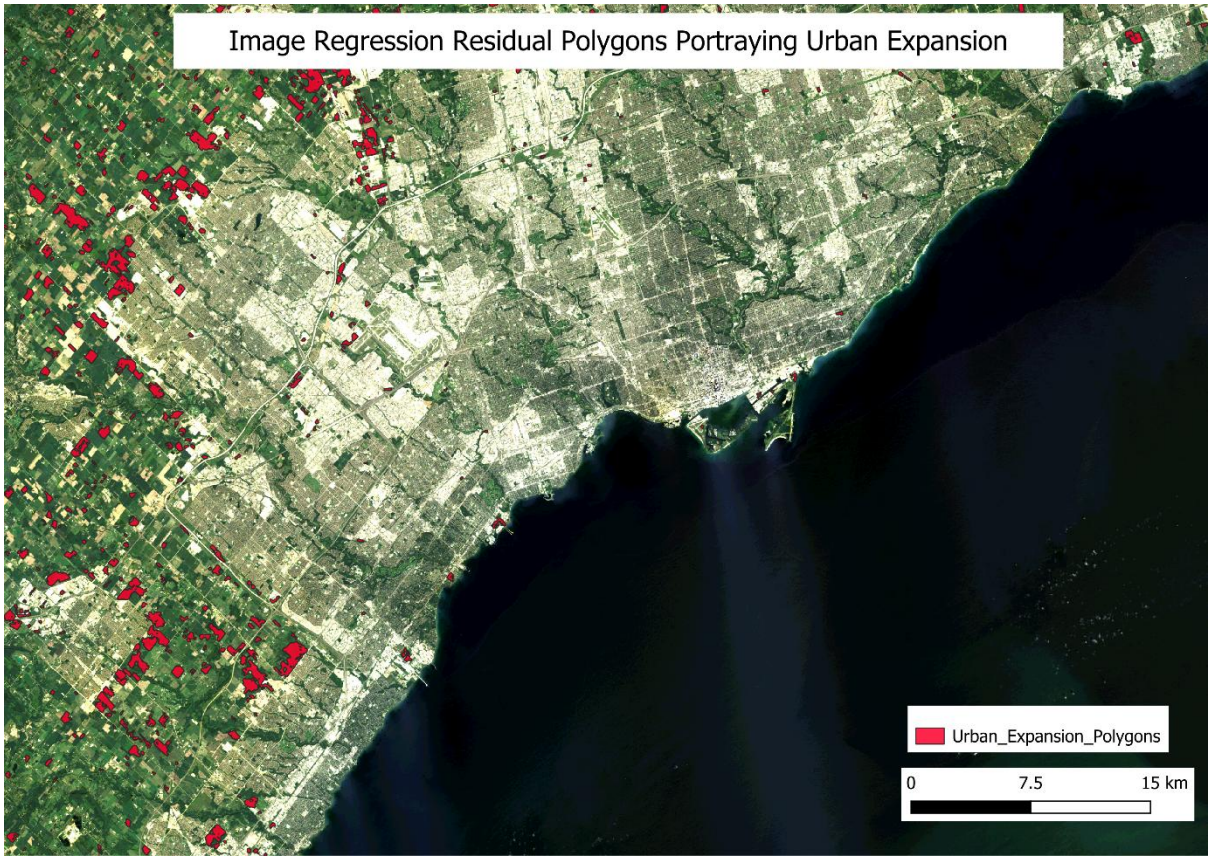

Figure 7. Regression residual map highlighting areas of significant vegetation decline between 2014 and 2024. Red polygons represent pixels with large negative residuals (greater

than 2.5× standard deviation), indicating likely urban and industrial expansion. These changes are concentrated around Milton, Brampton, and the western GTA. Polygons are overlaid on the RGB composite for spatial context.

The linear regression portrays a strong relationship between NDVI values across dates. Negative residuals highlight areas where vegetation declined more than expected. After thresholding and filtering, clusters of decline were exported as polygons. The map depicts the polygons overlaid on the 2014 RGB colour composite, which explains the growth of industrial and urban areas at the end of the ten-year period.

The northern parts of Oakville have seen urban growth, while Milton has seen industrial development along its outskirts. The northern parts of Brampton have seen industrial growth, though a few areas to the north are mainly farmland incorrectly classified as urban growth; some of these patches are merely changes in vegetation. Markham, Pickering, and Ajax have seen significant industrial expansion.

3) Random Forest Classification

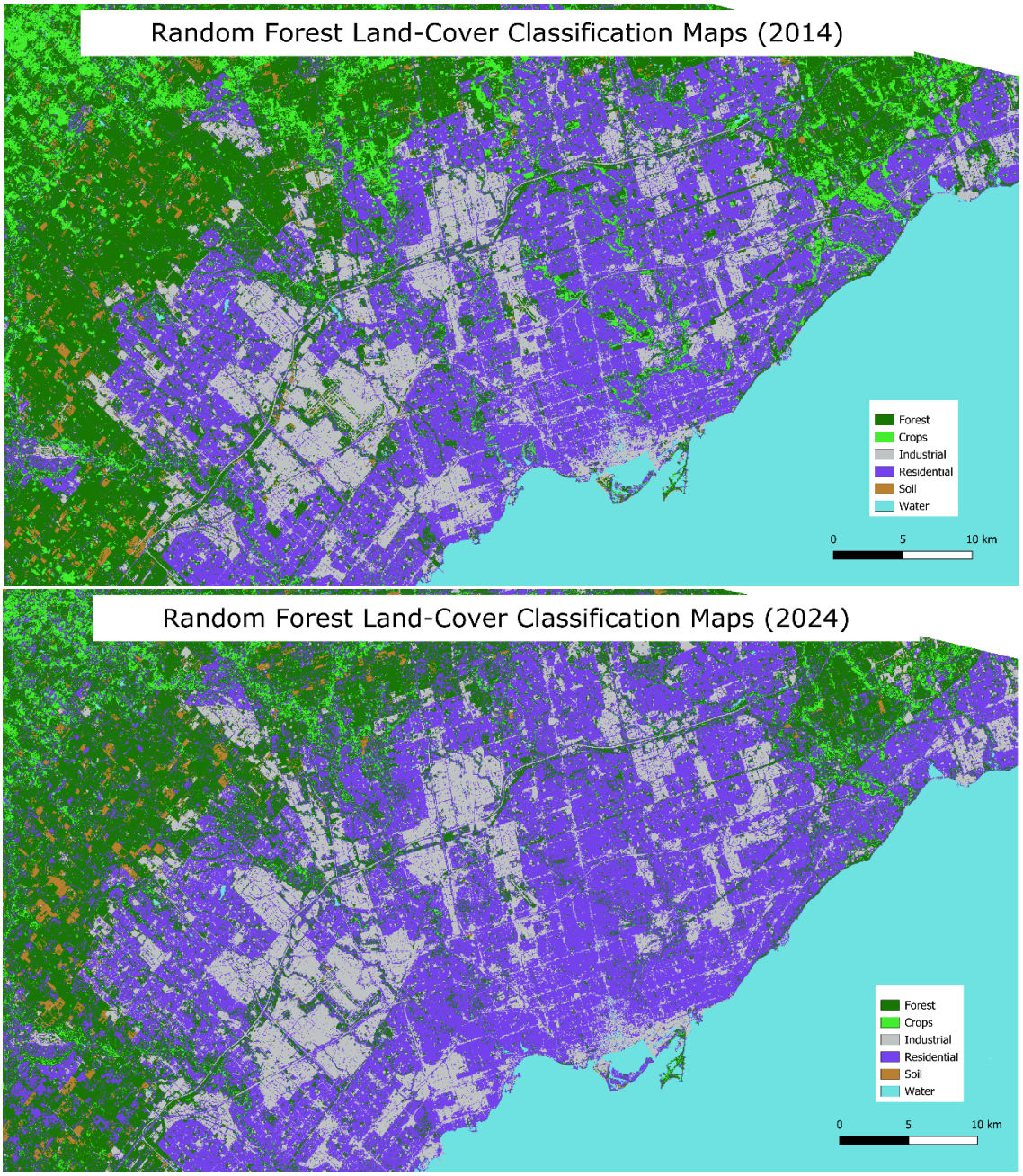

Figure 8. Random Forest classified land-cover maps for 2014 and 2024 after applying a 5×5 majority filter. Different colours represent land-cover classes.

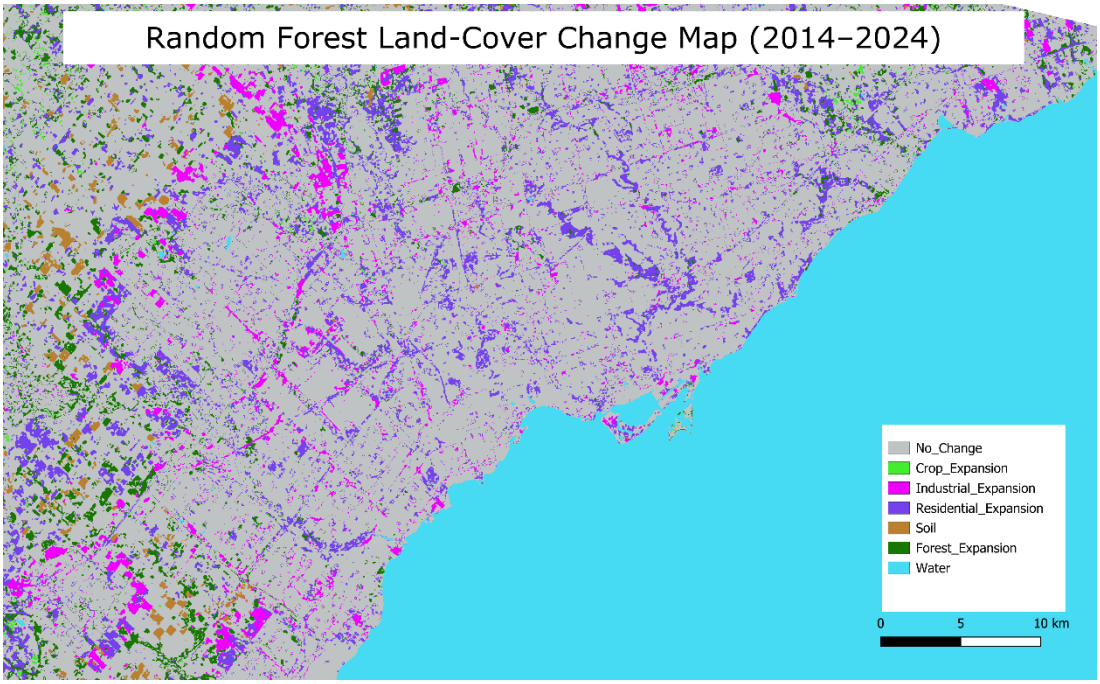

Figure 9. Post-classification change map derived from Random Forest analysis showing land-cover transitions from 2014 to 2024. Each coloured pixel represents a change to a specific land-cover type: purple indicates areas that have become residential, etc. Grey areas indicate no change between the two dates.

Cross-tabulation matrix (2014 → 2024)

Random Forest–classified land cover maps (5×5 majority filter applied)

2024

2014

	Forest	Industrial	Crops	Residential	Soil	Water
Forest	1,089,715	129,273	15,562	29,703	34,126	1,980
Industrial	14,632	96,895	7	794	112	322
Crops	87,928	568	442,427	40,939	3,416	838
Residential	201,569	73,886	66,453	1,105,718	4,257	4,503
Soil	47,960	297	653	264	5,278	1
Water	893	244	2,368	864	0	2,918,388

Table 2. Cross-tabulation matrix generated by comparing the 5×5 majority-filtered Random Forest classified maps for 2014 and 2024. Rows represent the original land-cover class (2014), columns represent the new land-cover class (2024), and each cell shows the total number of pixels that transitioned between those classes.

The RF classifications for 2014 and 2024 produce six class land cover maps. The 5 × 5 majority filter reduces the salt-and-pepper effect common in per-pixel classifiers. The cross-tabulation table provides a clear picture of the change that has occurred, although some aspects appear incorrect on inspection; this is likely due to misclassification by the algorithm.

129,000 forest pixels were converted to Industrial/commercial areas, depicting growth over the ten years. Cities and towns are expanding, and there is a greater need for more industrial areas. 29,000 Forest pixels were converted to residential areas, which is expected. Felling forests for farming could explain the 34,000-odd pixels being changed to soil cover.

40,000 crop pixels are being converted to residential pixels, depicting urban development and land loss. 48,000 soil pixels converted to forest pixels could indicate either forest or crop growth; both are possible given their similar pixel colour values.

15,000 industrial pixels are being converted to Forest pixels, which seems to be wrongly classified. It may be wrong to conclude that 88,000 Crop pixels are being converted to Forests, since both classes have similar pixel colours, which increases the risk of misclassification. The residential pixel changes seem unlikely and may be due to the poor quality of the training data used for that class.

Recommendations

Urban expansion between 2014 and 2024 is mostly seen in Milton, Oakville, Brampton, Markham, Pickering and Ajax. These fast-developing towns require investment in transportation networks, accessible grocery/convenience stores, sanitation and essential public services.

The significant conversion of forest and cropland pixels to industrial and residential land underscores the rising demand for housing and commercial space. Municipalities can use this information to make zoning decisions and attract the right investments to promote economic growth. The loss of forests and farmland, however, is an environmental concern and must be kept in check prior to conversion; a designated committee should oversee and approve decisions in this regard. Implementing green buffers, protecting remaining high-value farmland, and integrating sustainable development practices need to be a priority. Finally, since some classification errors were observed, the results should be validated before major decisions are made.

Limitations

NDVI is sensitive to seasonality; even if both images are from the same time of the month in different years, vegetation conditions may differ due to rainfall variation, crop growth, and drought, so changes may reflect phenological differences rather than actual land-use change (Mas, 1999). NDVI only measures vegetation greenness and cannot differentiate between types of change, such as residential vs industrial development, limiting its usefulness for detailed urban analysis (Jensen, 2016).

Image regression assumes a linear relationship between multiple dates of imagery, which may not hold in the real world, reducing the accuracy of detected change (Jensen, 2016). The identification of change relies on thresholds (e.g., 2.5× standard deviation), which are subjective and can significantly influence results (Mas, 1999).

Random Forest classification depends on the quality of the training data; poor samples per class can lead to incorrect classification results (Mather & Koch, 2011). Certain classes have similar reflectance values, which leads to spectral confusion and eventually misclassification, for example, cropland vs forests, residential areas with trees vs forests, bare soil vs construction sites. Errors in individual classified maps accumulate during comparison, reducing the reliability of detected change patterns (Mas, 1999).

These limitations are particularly relevant in this study, where NDVI and regression rely on vegetation signals, while Random Forest classification depends on training data and is affected by classification errors across both dates.

References

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